

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : ESC(CSBS) 502/PCC-AI 502/PCCAIML 502 Machine Learning

UPID : 005937

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What are outliers?
- (ii) Which of the following ensemble model helps in reducing variance?
- (iii) _____ is the machine learning algorithms that can be used with labeled data
- (iv) _____ is a part of machine learning that works with neural networks
- (v) Where does the Hidden Markov Model is used?
- (vi) How many types of Artificial Neural Networks?
- (vii) What is Auto Association Task?
- (viii) What are independent variables?
- (ix) Suppose you have picked the parameter for a model using 10-fold cross validation (CV). Which of the following is the best way to pick a final model to use and estimate its error?
- (x) The model will be trained with data in one single batch is known as ?
- (xi) Which suggests the existence of an efficient recursive algorithm for online smoothing?
- (xii) _____ stands for Statistical Information Grid.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What are the different types of clustering algorithms available? Which one would you recommend for a given problem? [5]
3. What is Overfitting ? How can you avoid it? [5]
4. What is the role of the Activation functions in Neural Networks? [5]
5. Why SVM is an example of a large margin classifier? [5]
6. Compare SVM and Logistic Regression in handling outliers. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Explain Perceptron. [5]
(b) what do you mean by Neural Network. [5]
(c) How do you choose the Scaling method used for Neural Networks? [5]
8. (a) What is linear regression? [4]
(b) State the assumptions in a linear regression model. [6]
(c) What is feature engineering? How do you apply it in the process of modelling? [5]
9. (a) Explain Random Forest method with suitable example. [7]
(b) Explain K-Means Clustering algorithm with suitable example. [8]
10. (a) How do you select important variables while working on a data set? [6]
(b) There are many machine learning algorithms till now. If given a data set, how can one determine which algorithm to be used for that? [6]
(c) What is the main key difference between supervised and unsupervised machine learning? [3]
11. (a) What do you mean Model Averaging? [5]
(b) How to Average Models? Explain with suitable example. [10]

*** END OF PAPER ***